Islamic University of Technology

Board Bazar, Gazipur, Dhaka

SWE 4304 - Software Project Lab - I

Project Proposal Report for **TEAM - 18**

LabTrack

Research Lab Management System

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SPL Team - 18

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01 : PROJECT OVERVIEW & MOTIVATION

1.1 OVERVIEW :

NAME : LabTrack

The Research Lab Management System (LabTrack) is a terminal-based application designed to organize and manage daily operations of a research laboratory. The system helps manage experiments, samples, protocols, inventory, equipment reservations, and lab room bookings in a centralized and structured manner.

The system supports multiple users with different roles, such as researchers, lab managers, technicians, and administrators. Each role has specific permissions, ensuring secure access and proper responsibility distribution.And based on their role, they can edit the experiments in a version control system and they can see the histories of the edits.

All operations are performed through a command-line interface, and all data is stored in text files, making the system lightweight, portable, and suitable for academic and small-scale lab environments.

1.2 MOTIVATION :

Modern research labs often rely on manual record-keeping or loosely organized digital files, which leads to inefficiency and errors. Through this project, we aim to address several common challenges faced by research laboratories.

This project will improve lab efficiency, accuracy, and accountability by providing a centralized system that:

1. Maintains structured records
2. Tracks changes using version control concepts
3. Prevents booking conflicts
4. Reduces human errors
5. Improves workflow transparency

02 : KEY FEATURES

The Research Lab Management System provides a set of integrated features that streamline laboratory operations and improve data organization. The system enables secure multi-user access with role-based permissions, ensuring that users can only perform actions relevant to their responsibilities. It supports experiment logging with version control, allowing changes to be tracked and previous versions to be reviewed. The system also manages laboratory protocols, tracks sample lineage, monitors inventory levels, and handles equipment and lab room reservations. Together, these features reduce manual errors, prevent resource conflicts, and ensure accurate, traceable, and efficient laboratory management.

2.1 Secure User Management :

The system supports multiple users with individual login accounts and role-based access control. Each user is assigned a role such as researcher, lab manager, technician, or administrator, which determines the actions they are allowed to perform. This prevents unauthorized access to sensitive operations and ensures accountability. The feature enhances system security while maintaining a clear division of responsibilities.

- **Username and Password Authentication :**
  - Users log in using valid credentials (Name, ID, Password).
  - Passwords are securely hashed before being stored in the csv files.

- **Role-Based Access Control :**
  - Researcher: Logs experiments and uses equipment
  - Lab Manager: Approves protocols and manages resources
  - Technician: Updates inventory and maintenance logs
  - Admin: Full system access

2.2 Experiment & Version Control :

The system allows users to create, update, and view experiment records in an organized manner. Each modification is saved as a new version, enabling users to track changes over time. Previous versions can be reviewed or restored when needed, ensuring data integrity. This feature helps maintain accurate experimental records and supports reproducibility.

- Create and update experiment records.
- Maintain multiple versions of experiments.
- View previous versions, restore if needed
- Track histories of who made changes.
- This feature is inspired by Git-Style version control to prevent data loss.

2.3 Protocol Management :

Standard operating procedures (SOPs) can be stored, updated, and managed within the system. Lab managers are responsible for approving or modifying protocols to ensure compliance with laboratory standards. The system maintains version history for protocols, allowing users to track changes. This feature ensures consistent and approved procedures are followed during experiments.

- Store Standard Operating Procedures (SOPs)
- Managers can approve, edit, or update protocols
- Version history maintained for protocol changes

Helps ensure standardized and approved lab practices.

2.4 Sample Lineage Tracking :

The system tracks the relationship between parent samples and derived samples throughout the experiment lifecycle. Users can trace a sample's origin, transformations, and usage across experiments. This improves traceability and research reliability. The feature is essential for maintaining accurate experimental documentation.

- Track parent and child samples
- Maintain full genealogy of samples
- Trace sample origin to final usage

This feature helps to boost research accuracy and reproducibility.

2.5 Inventory Management :

Inventory items such as reagents and materials are monitored with updated stock levels. Technicians can log usage, updates, and maintenance activities. The system helps prevent shortages and reduces material wastage. This feature ensures efficient resource utilization within the laboratory.

- Track real-time quantity of reagents
- Log material usage per experiment
- Generate low-stock alerts
- Technician-managed inventory updates

2.6 Equipment & Lab Reservation :

The system allows users to reserve laboratory equipment and rooms for specific time periods. It prevents scheduling conflicts by validating availability before confirming bookings.Usage and maintenance logs are maintained for accountability. This feature improves resource planning and operational efficiency.

- Reserve equipment for specific time slots
- Prevent booking conflicts
- Track equipment availability
- Maintain maintenance logs

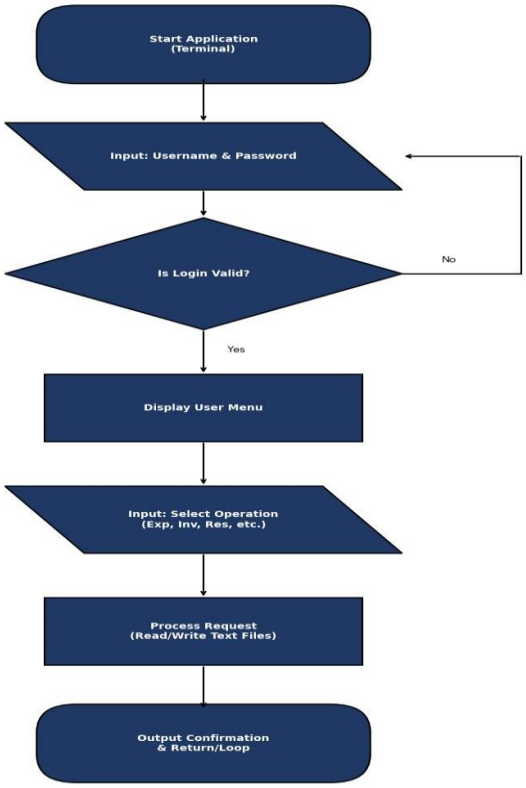
Also includes :

- Lab room booking
- Daily room usage logs
- History of experiments conducted in rooms

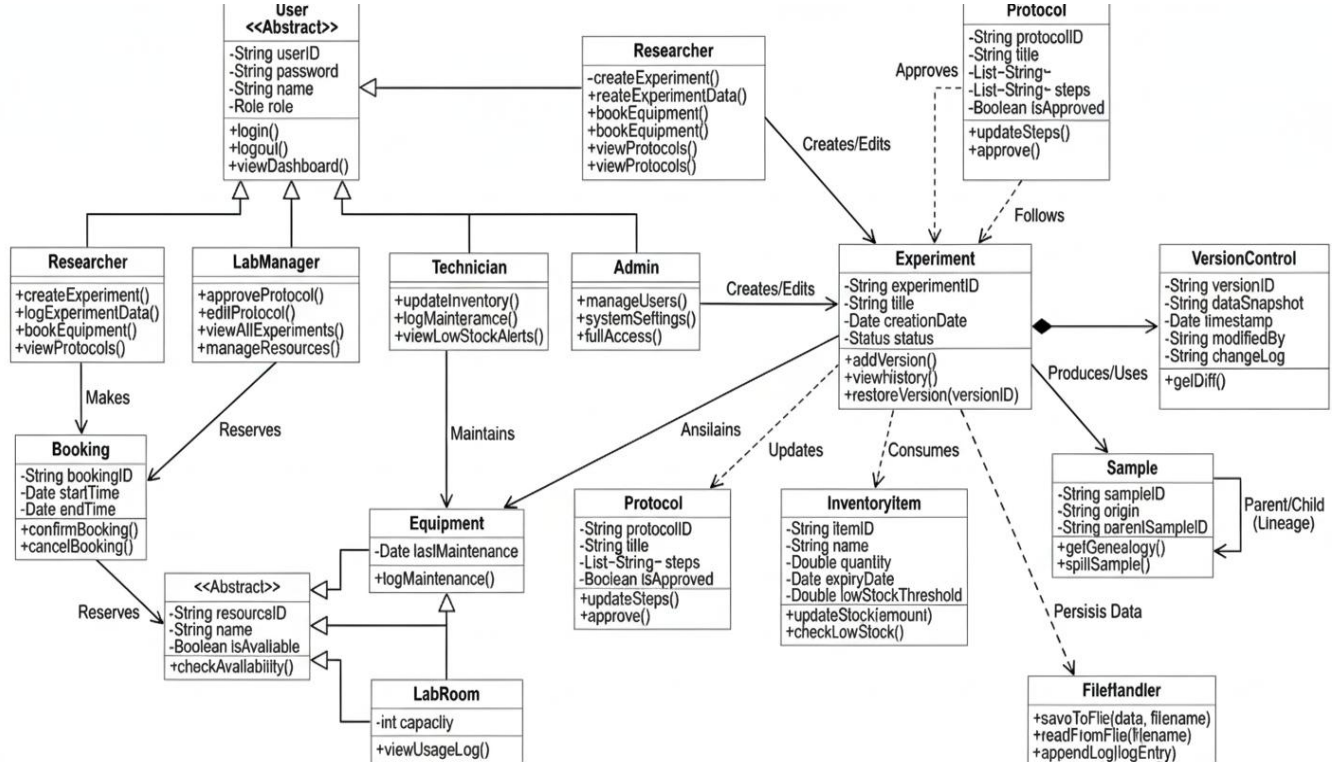
03 : FLOWCHART & CLASS DIAGRAM

3.1 System Workflow :

1. The system starts in the terminal.
2. The user logs in using a username and password.
3. Based on the user role, a menu is displayed.
4. The user selects an operation (experiments, inventory, reservations, etc.).
5. The system processes the request.
6. Data is stored or retrieved from text files.
7. The user receives confirmation or output.



3.2 UML Class Diagram :



04 : TOOLS & TECHNOLOGIES

4.1 Programming Language : JAVA

Java is used to develop the system due to its strong support for object-oriented programming and file handling. It enables the creation of a reliable and platform-independent terminal-based application.

4.2 Development Environment : VScode, IntelliJ IDEA

The project is developed using Visual Studio Code or IntelliJ IDEA, which provide efficient tools for coding, debugging, and testing. These environments help maintain clean and organized source code.

4.3 User Interface : Terminal

The system operates through a command-line interface, allowing users to interact using text-based menus. This approach keeps the application lightweight and suitable for environments without graphical interfaces.

4.4 Data Storage : Text File

All system data is stored in structured text files such as CSV and TXT formats. This method ensures simplicity, easy portability, and compliance with project constraints without using databases.

4.5 Version Control : Git

Git is used during development to manage source code versions and track changes. It helps maintain code consistency and supports collaborative development among team members.

4.6 Operating System : Windows, Linux

The application runs on Windows and Linux operating systems using the terminal. This ensures flexibility and ease of execution across different platforms.

4.7 Testing Method : Sample Test Datasets

Manual testing is performed using sample datasets to verify system functionality. This helps ensure correct behavior of features such as login, data storage, and module interactions.

05 : PROPOSED TIMELINE

WEEKS	ACTIVITIES
Week 2	Group formation
Week 3	Supervisor selection & draft proposal
Week 4	Proposal submission & presentation
Week 5	Finalize modules, class structure, project setup
Week 6	Core framework implementation (auth, file I/O, utilities)
Week 7	Version control, protocol & inventory modules
Week 8	Equipment tracking, maintenance, lab booking, samples
Week 9	Module integration & presentation
Week 10	First development phase
Week 11-13	Full development phase
Week 14	Final presentation, source code & report submission

5.1 System Design and Project Setup (Week 5) :

In this phase, the overall system architecture will be designed, and the project modules will be finalized. The class structure will be defined using object-oriented principles. The project folder structure, packages, and base classes will be set up to ensure maintainability and scalability.

5.2 Core Framework Development (Week 6) :

This phase will focus on building the foundation of the system. Custom string utilities, basic data structures, and a file input/output wrapper will be implemented. A secure user authentication system with role-based access control will also be developed to enable controlled multi-user interaction.

5.3 Advanced Module Development (Week 8) :

Additional modules will be developed to extend the system's capabilities. These will include equipment tracking, maintenance logging, laboratory room booking, and sample lineage tracking. These features will improve resource management and operational efficiency.

5.4 System Integration and Testing (Week 9) :

All developed modules will be integrated into a single functional system during this phase. Functional testing will be conducted to ensure smooth interaction between components. Any issues identified will be resolved, and the system will be prepared for presentation.

5.5 Final Development and Submission (Weeks 10–14) :

The final phase will focus on system refinement, performance improvement, and documentation. Additional testing and bug fixes will be performed to enhance reliability. The final report, source code, and presentation materials will be completed and submitted.

06 : ACTIONS TAKEN BASED ON FEEDBACK

6.1 Equipment Categorization, Access level and Borrowing :

The equipments should have categories based on use cases and who can access those equipments. Currently the project does not include this plan. But we are thinking how to add this feature. We will take feedbacks from our seniors and some faculty members to know more about how it will be efficient to implement the feature and then we will come up with the implementation process.

6.2 Too Much Text and Inconsistent Slide Numbering :

- In our slide presentation, there was a mistake from our side which was numbering the slide. At first we forgot to number the slides but at the very last moment we remembered the issues and so it was inconsistent. But we have already consistently numberized our slides now.
- Another issue was that our slide had too many texts. So we have simplified the texts and make the important notes bold so that in can be seen quickly.